

Content QA — processed GACAR corpus vs. official source

SECTION 06-operations-it TYPE document STATUS **active** OWNER Founder UPDATED 2026-06-16

The regulation corpus in `assets/data/parts/` is machine-extracted from GACA's published PDFs. This note records a spot-check of that extraction against the **official source** (<https://gaca.gov.sa/>) and what it found, so the result is reproducible and the next person doesn't have to redo it.

Method

1. `npm run check:content` triages every Part/library doc for extraction defects (`scripts/check-content.js`).
2. For high-stakes numeric rules, the processed `assets/data/parts/part-N.html` was compared verbatim against the official GACAR PDF for that Part, downloaded from gaca.gov.sa and text-extracted. The official Part PDFs live under:
`https://gaca.gov.sa/-/media/Files/PDF/LawsAndRegulation/Aviation-Safety-and-Environmental-Sustainability/GACAR-Safety-Regulations/` (e.g. `Part-091---General-Operating-and-Flight-Rules.pdf`).

Result — numbers are accurate

Every regulatory figure checked matches the official source exactly, **including the Saudi-specific values that differ from the FAA baseline** the content is modelled on:

Rule	Official GACAR	Processed corpus	Verdict
§91.65 max IAS < 10,000 ft MSL	250 kt (463 km/h)	250 kt (463 km/h)	✓
§91.65 max IAS near Class C/D, under Class B	200 kt (370 km/h)	200 kt (370 km/h)	✓
Supplemental O ₂ — crew, entire flight, above	13,000 ft MSL (not FAA 14,000)	13,000 ft MSL	✓
Supplemental O ₂ — pressurized, mask above	FL 250 (not FAA FL 350)	FL 250	✓
§91.155 VFR vis, Class B / ≥10,000 ft	5 KM clear of cloud / 8 KM, 1,500 m	same	✓
Part 61 PPL minimum age (non-glider/balloon)	17 years	17 years	✓

Conclusion: the corpus is **substantively trustworthy**. No numeric corrections were needed from this spot-check.

Defects found — presentation, not content

The extraction left cosmetic scarring that does **not** change the regulatory numbers but hurts readability and, in one case, navigation. `scripts/check-content.js` now detects each class (run it for the current per-file list; `--strict` exits non-zero on structural defects):

- **Math-OCR scarring** (53/74 Part files, ~1,000 hits): bold section numbers were read as LaTeX, leaving `mathbf$91.33`, `bfxi91-19`, `bfdeltadelta91-21`. This also **mangles the** `id="sec-..."` **anchors** that `assets/js/document.js` deep-links to (a link to `#sec-91-41` misses the actual id `sec-bf-91-41`) — the highest-impact defect.
- **Stray `<html>/<body>` table wrappers** (32 files): extracted reg tables wrapped as standalone documents inside `<figure>`; `innerHTML` strips the wrappers so they still render, but cell text carries OCR noise (`1000 f` for "ft").
- **Duplicate `id` s** (29 files): repeated `sub-a / sub-f` etc. break deep-link targeting.
- **Char-level OCR** spotted in spot-checks: `4o` → 40, `3e` → Be, lost spaces (`2 500ft`), `ICAO`, `rajjectory`, `ACAS IHI`.

Recommendation

Content accuracy is good enough to ship. The remaining work is a **presentation cleanup**, best done by re-ingesting from the now-located source PDFs with a math-mode-aware extractor (or a targeted transform that strips the `mathbf/bf*` tokens and regenerates `sec-...` ids). Track progress with `node scripts/check-content.js --strict` — it goes green when a Part is clean.